The University of New South Wales

School of Electrical Engineering and Telecommunications

ELEC3004: Signal Processing and Transform Methods

Final Examination, Session 2, 2004

Candidates should attempt ALL questions

1. Time allowed: **3** hours.

2. The paper contains **4** questions.

3. The questions are NOT of equal value. Part marks are as indicated .

4. The examination paper may **not** be retained by the candidate.

5. Casio $f\chi$ - 911W scientific calculator provided by examination section.

6. Transform tables are supplied for your use.

ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK.

Question 1. (45 marks)

A. (i) Determine the transfer function $H(z)$ of a discrete-time system with the pole-zero diagram shown in Figure 1. **[2 marks]**

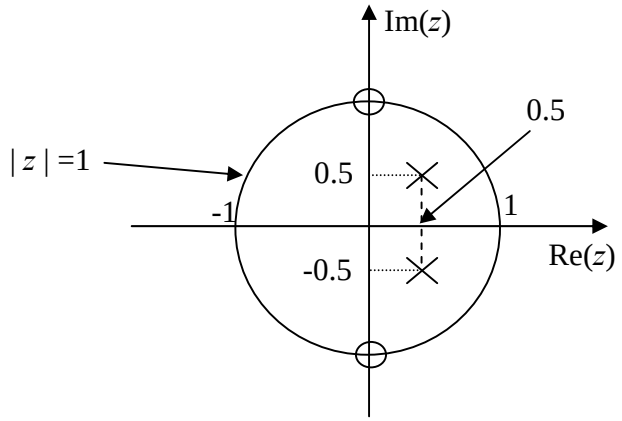


Figure 1

(ii) Sketch the approximate magnitude response for the pole-zero map given in Figure 1 **[3 marks]**

B. Suppose $x(t)$ has the spectrum $X(f)$ as shown in Figure 2. Sketch the digital spectrum $|H(\theta)|$ if the sampling frequency f_s = 2 kHz.

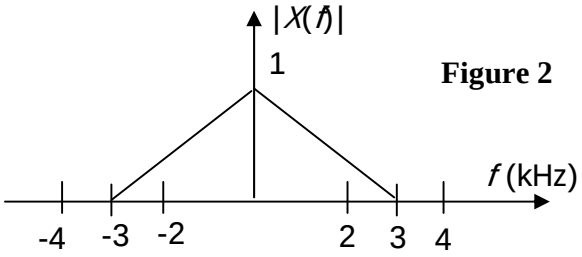


Figure 2

[4 marks]

C. Consider the lowpass filter

$$y[n] = a\,y[n-1] + b\,x[n], \qquad 0 < a < 1$$

(i) Determine b such that $|H(0)| = 1$. **[1 mark]**

(ii) Determine the 3-dB bandwidth θ_c for the normalised filter in part (i). **[4 marks]**

D. The transfer function for a frequency-sampling filter is given by

$$P(z) = \frac{1-z^{-N}}{N}\left[\frac{H_0}{1-z^{-1}} + \left\{\frac{H_1}{1-z^{-1}e^{j2\pi/N}} + \frac{H_{-1}}{1-z^{-1}e^{-j2\pi/N}}\right\}\right]$$

For a real filter $H_{-1} = H_1^*$ (i.e. complex conjugate). Assume N =3, H_0 = 3 and $H_1 = 3(2 + j\sqrt{3})$.

(i) Draw the frequency-sampling filter structure using delay elements, multipliers and adders. **[5 marks]**

(ii) From the results obtained in part (i) above, $P(z)$ can be realised as an FIR filter. Find the coefficients of the FIR filter. **[3 marks]**

E. A transversal filter has seven filter coefficients b_0 to b_6, with $b_0 = 1$, $b_1 = 2$, $b_2 = 3$ and $b_3 = 4$.

(i) Determine b_4 to b_6 such that this filter has linear phase characteristics. **[3 marks]**

(ii) If all the coefficients (b_0 to b_6) were set to unity, show that the filter has linear phase characteristics and is given by $\phi(\theta) = -3\theta$. Draw its amplitude characteristics ($|H(\theta)|$ vs θ). **[5 marks]**

(iii) Show that the above filter has a constant group delay. **[2 marks]**

F. A digital filter is represented by

$$H(z) = \frac{\frac{1}{2}}{1-\frac{1}{2}z^{-1}} + \frac{1}{(1-\frac{1}{2}z^{-1})(1-a\,z^{-1})}$$

(i) Does this transfer function represent an FIR or an IIR filter? **[1 mark]**

(ii) Write a difference equation for $H(z)$ using direct form. **[2 mark]**

(iii) Implement a parallel realisation of $H(z)$. **[3 mark]**

(iv) Determine the impulse response of the above digital filter $H(z)$ and show that the filter is stable if $|a| < 1$ **[3 mark]**

G. The transfer function of a digital filter is given by

$$H(z) = \frac{3 + 3z^{-1}(e^{-2T} - 2e^{-T})}{(1-e^{-2T}z^{-1})(1-e^{-T}z^{-1})}$$

where T is the sampling period.

Assume that $H(z)$ was designed using the impulse invariant transformation method. Find the analogue transfer function $H(s)$. **[4 marks]**

Question 2. (15 marks)

A. The transfer function of a digital oscillator is given by

$$H(z) = \frac{z}{z^2 - 2\cos(\theta_0)z + 1} = \frac{z}{z^2 - b_1 z + 1}$$

(i) Show that the impulse response corresponding to *H(z)* is given by

$$h[n] = \frac{\sin[n\theta_0]\quad u[n]}{\sin[\theta_0]}$$ **[2 Marks]**

(ii) The frequency of oscillation of the system $H(z)$ can be determined by the coefficient b_1. Show that

$$\Delta f_0 = \frac{-f_s\ \Delta b_1}{4\pi \sin(2\pi \frac{f_0}{f_s})}$$

where f_S = sampling frequency and f_O = desired frequency of oscillation. **[2 Marks]**

(iii) Calculate the highest frequency resolution(Δf_o) that is obtainable for this oscillator. **[2 Marks]**

B. Another form of an oscillator is given by the following coupled difference equations expressed in matrix form:

$$\begin{bmatrix} y_C[n] \\ y_S[n] \end{bmatrix} = \begin{bmatrix} \cos\theta_0 & -\sin\theta_0 \\ \sin\theta_0 & \cos\theta_0 \end{bmatrix} \begin{bmatrix} y_C[n-1] \\ y_S[n-1] \end{bmatrix}$$

(i) Draw the structure for the realisation of this oscillator, where θ_0 is the oscillation frequency. **[3 Marks]**

(ii) If the initial conditions $y_c[-1] = A\cos\theta_0$ and $y_s[-1] = -A\sin\theta_0$, Show that the output $y_c[n] = A\cos[n\theta_0]$. **[3 Marks]**

C. Consider the following causal IIR transfer function:

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2 + z + 0.5)}$$

Is $H(z)$ a stable function? If it is not stable, find a stable transfer function $G(z)$ such that $|G(\theta)| = |H(\theta)|$.

[3 Marks]

Question 3. (25 marks)

A. A digital filter is required to meet the following specifications:

Low-pass filter:	0 to 10 kHz passband
Sampling frequency:	f_s = 100 kHz
Transition band:	10 kHz to 20 kHz
Stopband attenuation :	-10 dB(Starting at 20 kHz)

The filter is to be designed by performing a bilinear transformation on an analogue filter. The filter must be monotonic in the pass and stop bands (i.e. no ripples).

(i) Determine the order of the analogue filter that must be used to meet the above specifications . **[5 marks]**

(ii) Obtain the transfer function, $G(z)$ of the digital filter and show that $G(z)$ reduces to

$$G(z) = 0.0675\frac{(1+2.0\,z^{-1}+z^{-2})}{1-1.143z^{-1}+0.4123z^{-2}}$$ **[6 marks]**

(iii) Draw a canonic realisation of $G(z)$. **[2 marks]**

B. You are required to design a linear phase FIR filter. The length of the impulse response is set to N (= 7) and the coefficients are to be determined using a windowing technique. An ideal lowpass filter $H_d(\theta)$, with linear phase can be characterised in the frequency domain by

$$H_d(\theta) = \begin{cases} e^{-j\beta\theta} & 0 \le |\theta| \le \dfrac{\pi}{7} \\ 0 & \text{otherwise} \end{cases}$$

(i) Determine β. **[2 marks]**

(ii) Determine the impulse response $h_d[n]$. **[5 marks]**

(iii) Calculate the coefficients of the FIR filter using the Bartlett window. **[5 marks]**

Note: The Bartlett Window function is given by

$$w_B[n] = \begin{cases} \dfrac{2n}{N-1} & 0 \le n < \dfrac{(N-1)}{2} \\ 2-\dfrac{2n}{N-1} & \dfrac{N-1}{2} \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}$$

where N is the length of the window.

Question 4. (15 marks)

A. Figure 3 below shows a section of a digital modulator. If the input $B_C(\theta) = 1$ for $-\pi \le \theta \le \pi$,

(i) Write an equation for $R_C(\theta)$, where $\theta_1 = 3\pi/4$. **[4 marks]**

(ii) Sketch $|R_C(\theta)|$ vs θ, **[4 marks]**

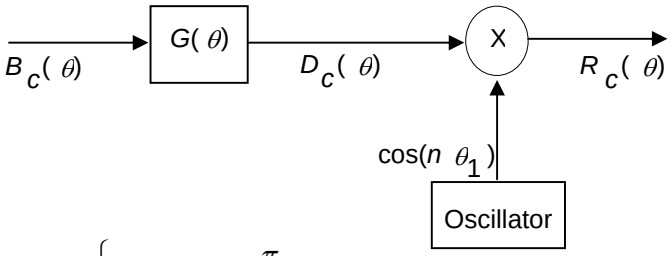


Figure 3

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \dfrac{\pi}{4} \\ 0 & \dfrac{\pi}{4} < |\theta| \le \pi \end{cases}$$

$$Note: \qquad x(n)\cos(\theta_0 n) \quad \Leftrightarrow \quad \frac{1}{2}\left[X(\theta-\theta_0)+X(\theta+\theta_0)\right]$$

B. Express the output y[n] of the figure below as a function of the input x[n]. By simplifying the expression, show that y[n] = x[n-1].

[7 marks]

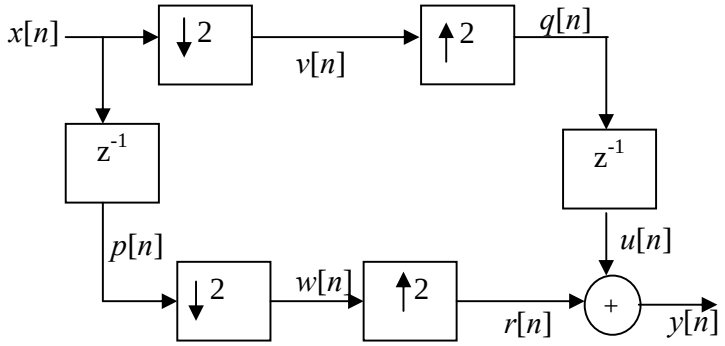
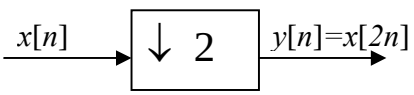
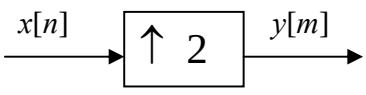


Figure 4

Note:



$$Y(z) = \frac{1}{2}\left[X(z^{\frac{1}{2}}) + X(-z^{\frac{1}{2}})\right]$$



$$Y(z) = \left[X(z^2)\right]$$

Table of the z transforms of useful sequences

Basic z-Transforms	**Sequence**	**z-Transform**
1. Unit sample	$\delta[n]$	1
	$\delta[n-k]$	z^{-k}
2. Unit step	u[n]	z/(z-1)
3. Exponential	a^n u[n)]	z/(z-a)
4. Sinusoidal	$\sin[\theta_o n]$ u[n]	$\frac{z \sin\theta_0}{z^2 - 2\cos\theta_0\ z + 1}$
	$\cos[\theta_o n]$ u[n]	$\frac{z^2 - z\cos\theta_0}{z^2 - 2\cos\theta_0\ z + 1}$
5. Unit ramp	nu[n]	$\frac{z}{(z-1)^2}$

$$1 + z^{-1} + z^{-2} + z^{-3} + \ldots\ldots\ldots + z^{-(N-1)} = \frac{1 - z^{-N}}{1 - z^{-1}}$$

z-Transform Properties	**Sequence**	**z-Transform**
	x[n]	X(z)
	x[n-k]	z^{-k} X(z)
	a^n x[n]	X(z/a)
	x[-n]	$X(z^{-1})$
	x[n]*y[n]	X(z).Y(z)
	n x[n]	-z (dx(z)/dz)

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\cos\theta = \frac{1}{2}\left[e^{j\theta} + e^{-j\theta}\right]$$
$$\sin\theta = \frac{1}{2j}\left[e^{j\theta} - e^{-j\theta}\right]$$

Fourier Series:

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t); \qquad \omega_0 = 2\pi f_0; \quad f_0 = \frac{1}{T}$$

The fundamental frequency of x(t) is ω_0 and T is the period of the signal x(t).

$$A_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(n\omega_0 t)\,dt \qquad n = 0,1,2.... \qquad B_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(n\omega_0 t)\,dt \qquad n = 1,2,3...$$

Fourier Transform pair (for discrete-time signal)

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}... \qquad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

$\theta = digital\ frequency; \quad -\pi \le \theta \le \pi$

A **Butterworth filter** of order'n' is defined by

$$|H(j\omega)|^2 = \frac{1}{1+\left(\frac{\omega}{\omega_C}\right)^{2n}}$$ where ω_c is the cut-off frequency of the filter

n	Butterworth Polynomials
1	$S + 1$
2	$s^2 + \sqrt{2}s + 1$
3	$(s^2 + s + 1)(s + 1)$

Impulse Invariant Method:

$$\frac{1}{s+b} \rightarrow \frac{1}{1 - e^{-bT} z^{-1}} \qquad b > 0$$ *T*-sampling period

Bilinear Transformation:

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}} \qquad T - sampling\ \ period$$

End of paper